Annex II. Public Debt Sustainability Analysis

Table 1. Peru: Baseline Scenario

(in percent of GDP, unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

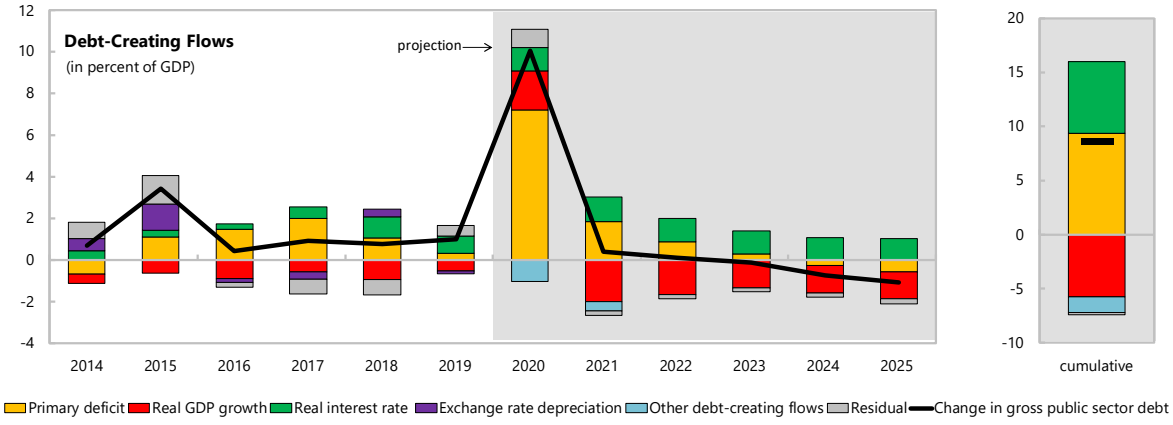
	Actual			Projections					
	2009-2017 [2/]	2018	2019	2020	2021	2022	2023	2024	2025
Nominal gross public debt	23.6	26.2	27.2	37.2	37.6	37.7	37.6	36.9	35.8
Of which: guarantees	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Public gross financing needs	2.3	3.8	2.3	9.2	3.9	4.0	4.4	4.8	4.0
Real GDP growth (in percent)	4.4	4.0	2.2	-6.5	5.8	4.7	3.7	3.7	3.7
Inflation (GDP deflator, in percent)	3.1	1.3	2.1	1.6	1.7	2.0	2.0	2.0	2.0
Nominal GDP growth (in percent)	7.8	6.0	3.8	-5.6	8.2	7.3	6.3	6.2	6.2
Effective interest rate (in percent) [4/]	5.2	5.6	5.5	5.4	5.3	5.3	5.1	5.1	5.0

As of May 06, 2020

Sovereign Spreads		
EMBIG (bp) 3/		236
5Y CDS (bp)		102
Ratings	Foreign	Local
Moody's	A3	A3
S&Ps	BBB+	A-
Fitch	BBB+	A-

Contribution to Changes in Public Debt

	Actual			Projections							
	2009-2017	2018	2019	2020	2021	2022	2023	2024	2025	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	-0.3	0.8	1.0	10.1	0.4	0.1	-0.1	-0.7	-1.1	8.6	
Identified debt-creating flows	-1.2	1.5	0.5	9.2	0.6	0.3	0.1	-0.5	-0.9	8.8	
Primary deficit	-0.5	1.0	0.3	7.2	1.8	0.9	0.3	-0.3	-0.6	9.4	-0.3
Primary (noninterest) revenue and grants	25.7	24.4	24.7	22.5	25.2	25.3	25.3	25.3	25.3	148.8	
Primary (noninterest) expenditure	25.2	25.4	25.1	29.7	27.0	26.2	25.6	25.0	24.7	158.2	
Automatic debt dynamics [5/]	-0.6	0.4	0.2	3.0	-0.8	-0.5	-0.2	-0.2	-0.3	0.9	
Interest rate/growth differential [6/]	-0.5	0.1	0.3	3.0	-0.8	-0.5	-0.2	-0.2	-0.3	0.9	
Of which: real interest rate	0.5	1.0	0.8	1.1	1.2	1.1	1.1	1.1	1.0	6.6	
Of which: real GDP growth	-1.0	-1.0	-0.5	1.9	-2.0	-1.7	-1.3	-1.3	-1.3	-5.7	
Exchange rate depreciation [7/]	-0.1	0.4	-0.1	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	0.0	0.0	-1.0	-0.4	0.0	0.0	0.0	0.0	-1.5	
General government net privatization proceeds (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Use of Stabilization Fund	0.0	0.0	0.0	-1.0	-0.4	0.0	0.0	0.0	0.0	-1.5	
Residual, including asset changes [8/]	0.9	-0.7	0.5	0.9	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	



Source: IMF staff calculations.

1/ Public sector is defined as non-financial public sector and includes public guarantees.

2/ Based on available data.

3/ EMBIG.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes changes in the stock of guarantees, asset changes, and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

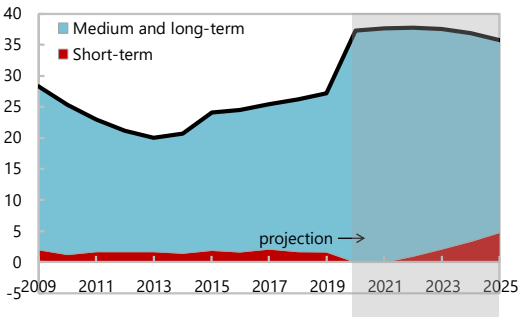
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Figure 1. Peru: Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

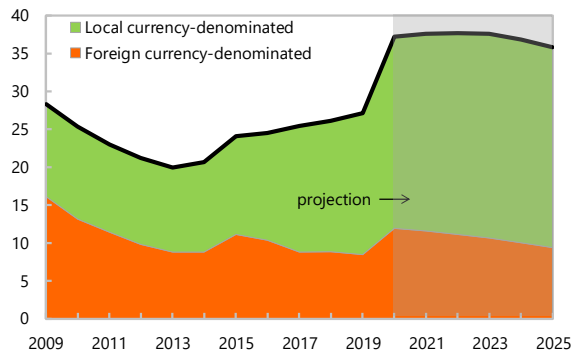
By Maturity

(in percent of GDP)

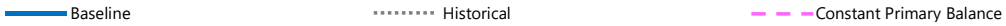


By Currency

(in percent of GDP)

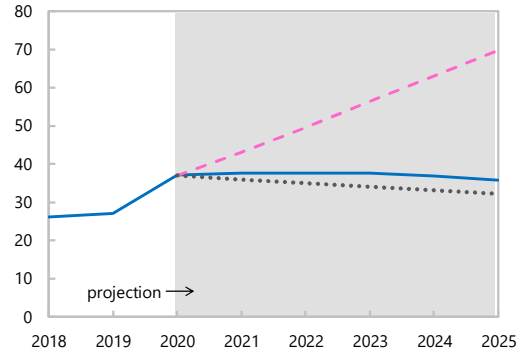


Alternative Scenarios



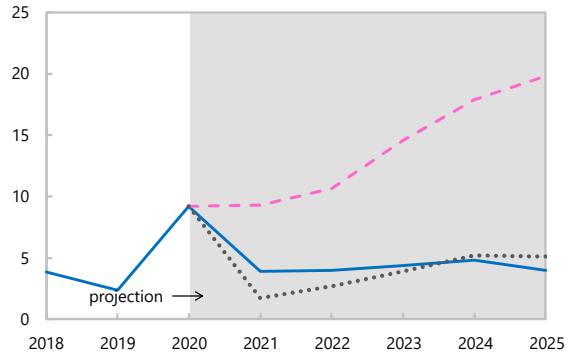
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-6.5	5.8	4.7	3.7	3.7	3.7
Inflation	1.6	1.7	2.0	2.0	2.0	2.0
Primary Balance	-7.2	-1.8	-0.9	-0.3	0.3	0.6
Effective interest rate	5.4	5.3	5.3	5.1	5.1	5.0
Constant Primary Balance Scenario						
Real GDP growth	-6.5	5.8	4.7	3.7	3.7	3.7
Inflation	1.6	1.7	2.0	2.0	2.0	2.0
Primary Balance	-7.2	-7.2	-7.2	-7.2	-7.2	-7.2
Effective interest rate	5.4	5.3	5.2	4.9	4.9	4.8

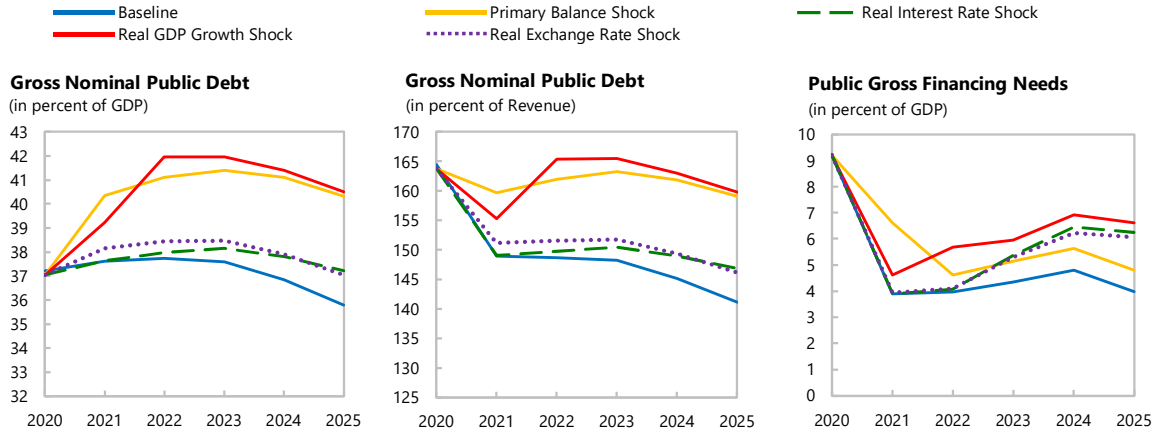
Historical Scenario	2020	2021	2022	2023	2024	2025
Real GDP growth	-6.5	4.5	4.5	4.5	4.5	4.5
Inflation	1.6	1.7	2.0	2.0	2.0	2.0
Primary Balance	-7.2	0.4	0.4	0.4	0.4	0.4
Effective interest rate	5.4	5.3	5.3	5.1	5.1	5.0

Source: IMF staff calculations.

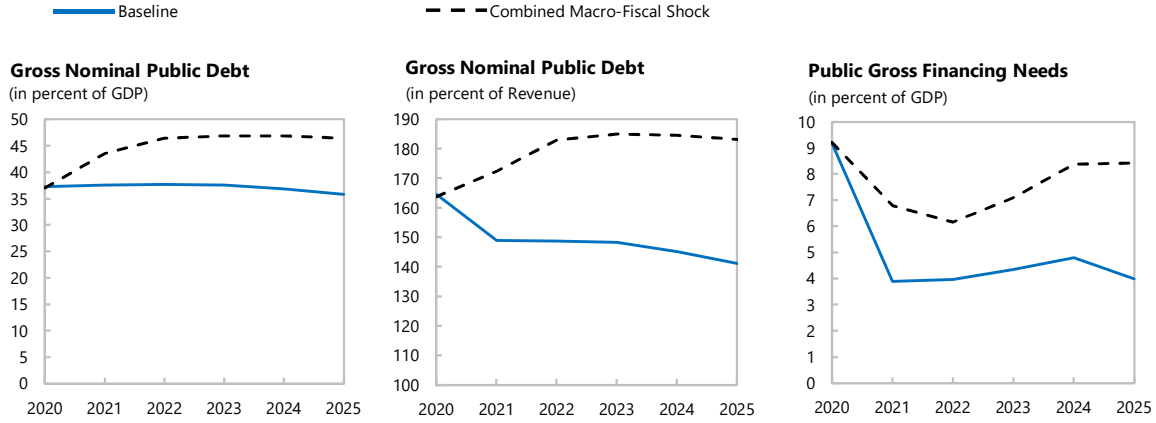
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Figure 2. Peru: Public Debt Stress Tests

Macro-Fiscal Stress Tests



Additional Stress Tests



Underlying Assumptions

(in percent)

Primary Balance Shock	2020	2021	2022	2023	2024	2025
Real GDP growth	-6.5	5.8	4.7	3.7	3.7	3.7
Inflation	1.6	1.7	2.0	2.0	2.0	2.0
Primary balance	-7.2	-4.5	-1.3	-0.6	0.0	0.4
Effective interest rate	5.4	5.3	5.3	5.2	5.1	5.0
Real Interest Rate Shock						
Real GDP growth	-6.5	5.8	4.7	3.7	3.7	3.7
Inflation	1.6	1.7	2.0	2.0	2.0	2.0
Primary balance	-7.2	-1.8	-0.9	-0.3	0.3	0.6
Effective interest rate	5.4	5.3	5.4	5.5	5.7	6.0
Combined Shock						
Real GDP growth	-6.5	3.7	2.6	3.7	3.7	3.7
Inflation	1.6	1.2	1.5	2.0	2.0	2.0
Primary balance	-7.2	-4.5	-2.3	-0.6	0.0	0.4
Effective interest rate	5.4	5.6	5.5	5.6	5.8	6.0

Real GDP Growth Shock	2020	2021	2022	2023	2024	2025
Real GDP growth	-6.5	3.7	2.6	3.7	3.7	3.7
Inflation	1.6	1.2	1.5	2.0	2.0	2.0
Primary balance	-7.2	-2.5	-2.3	-0.3	0.3	0.6
Effective interest rate	5.4	5.3	5.3	5.2	5.2	5.2
Real Exchange Rate Shock						
Real GDP growth	-6.5	5.8	4.7	3.7	3.7	3.7
Inflation	1.6	6.5	2.0	2.0	2.0	2.0
Primary balance	-7.2	-1.8	-0.9	-0.3	0.3	0.6
Effective interest rate	5.4	5.6	5.2	5.1	5.1	5.2

Source: IMF staff.

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